

# ME 492 — The Calculation Questions

Every question they ask about the tables, and how to answer it. Worked in full.

Section D is **6 to 9 questions** and there are only **three** table types. Learn the three methods and the section is finished in ten minutes. All questions carry equal marks, so these are the cheapest marks on the paper.

**The numbers change every year. The method never does.** The same tables appear under different company names: Topsy Turvy, Kaneapa, Dandy Electronics, GoldCom, Inky Pinky, Hokus Pokus, Akua Antwiwaah, Osagyefo, Birdy-Birdy's, Lovy Dovy, Mercy's Bookshop. Do not memorise a figure.

## 1. What they actually ask

These are the question wordings that appear, taken from the papers.

| Type | The wording                                                                                                          | How often   |
|------|----------------------------------------------------------------------------------------------------------------------|-------------|
| 1    | Determine the accounts receivable for October / November / December                                                  | common      |
| 1    | Determine the budgeted cash sales for October / November / December                                                  | common      |
| 1    | Determine the amount of accounts receivable that will appear on the company's fourth quarter pro forma balance sheet | every paper |
| 1    | Determine the amount of sales revenue that will appear on the company's fourth quarter pro forma income statement    | every paper |
| 1    | Compute the expected increase in sales per month                                                                     | rare        |
| 2    | Determine the sales in the fourth quarter for the [West Division / Accra District]                                   | every paper |
| 2    | Determine the amount of sales revenue on the fourth quarter pro forma income statement                               | every paper |
| 3    | Compute the total cash available for August / September                                                              | common      |
| 3    | Determine the cash shortage / cash surplus for August / September                                                    | common      |
| 3    | Calculate the amount to be borrowed / repaid for August / September                                                  | common      |
| 3    | Compute the total amount of interest expense for the third quarter                                                   | common      |
| 3    | Compute the ending cash balance for the month of September                                                           | common      |

## 2. Type 1 — Sales budget and schedule of cash receipts

You are given cash sales, sales on account and total budgeted collections for October, November and December, plus an opening accounts receivable balance at 1 October. The company collects **100 percent of accounts receivable in the month following the month of sale.**

### The five rules

1. **Accounts receivable for a month** = the credit sales of the **previous** month. For October, use the opening balance you were given.
2. **Budgeted cash sales for a month** = the cash sales line. Read it off the table. This is a free mark.
3. **Total budgeted collections** = this month's cash sales + last month's credit sales. Use it to check yourself.
4. **AR on the Q4 pro forma balance sheet** = **December's credit sales**. That is what is still owed at 31 December.
5. **Sales revenue on the Q4 pro forma income statement** = the sum of the three total budgeted sales figures.

## Topsy Turvy Company Ltd

|                                      | October         | November        | December        |
|--------------------------------------|-----------------|-----------------|-----------------|
| Cash sales                           | \$15,000        | \$17,250        | \$19,838        |
| Sales on account (credit)            | \$45,000        | \$51,750        | \$59,513        |
| <b>Total budgeted sales</b>          | <b>\$60,000</b> | <b>\$69,000</b> | <b>\$79,351</b> |
| Collections from accounts receivable | \$40,000        | \$45,000        | \$51,750        |
| <b>Total budgeted collections</b>    | <b>\$55,000</b> | <b>\$62,250</b> | <b>\$71,588</b> |

| The question                             | Working                                                 | Answer           |
|------------------------------------------|---------------------------------------------------------|------------------|
| AR for October                           | the given opening balance on 1 October                  | <b>\$40,000</b>  |
| AR for November                          | October credit sales                                    | <b>\$45,000</b>  |
| AR for December                          | November credit sales                                   | <b>\$51,750</b>  |
| Cash sales, October                      | read straight off the table                             | <b>\$15,000</b>  |
| Cash sales, November                     | read straight off the table                             | <b>\$17,250</b>  |
| Cash sales, December                     | read straight off the table                             | <b>\$19,838</b>  |
| AR on the Q4 balance sheet               | December credit sales, still uncollected at 31 December | <b>\$59,513</b>  |
| Sales revenue on the Q4 income statement | 60,000 + 69,000 + 79,351                                | <b>\$208,351</b> |

## Dandy Electronics Company Ltd

|                                      | October             | November            | December            |
|--------------------------------------|---------------------|---------------------|---------------------|
| Cash sales                           | GH¢787,500          | GH¢984,375          | GH¢1,230,469        |
| Sales on account (credit)            | GH¢337,500          | GH¢421,875          | GH¢527,344          |
| <b>Total budgeted sales</b>          | <b>GH¢1,125,000</b> | <b>GH¢1,406,250</b> | <b>GH¢1,757,813</b> |
| Collections from accounts receivable | GH¢0                | GH¢337,500          | GH¢421,875          |
| <b>Total budgeted collections</b>    | <b>GH¢787,500</b>   | <b>GH¢1,321,875</b> | <b>GH¢1,652,344</b> |

| The question                             | Working                                                 | Answer              |
|------------------------------------------|---------------------------------------------------------|---------------------|
| AR for October                           | the given opening balance on 1 October                  | <b>GH¢0</b>         |
| AR for November                          | October credit sales                                    | <b>GH¢337,500</b>   |
| AR for December                          | November credit sales                                   | <b>GH¢421,875</b>   |
| Cash sales, October                      | read straight off the table                             | <b>GH¢787,500</b>   |
| Cash sales, November                     | read straight off the table                             | <b>GH¢984,375</b>   |
| Cash sales, December                     | read straight off the table                             | <b>GH¢1,230,469</b> |
| AR on the Q4 balance sheet               | December credit sales, still uncollected at 31 December | <b>GH¢527,344</b>   |
| Sales revenue on the Q4 income statement | 1,125,000 + 1,406,250 + 1,757,813                       | <b>GH¢4,289,063</b> |

**Trap.** Rule 4 asks for December **credit** sales, not December total sales. The distractor is always the total. In Dandy that is GH¢527,344, not GH¢1,757,813.

**Trust the table, not the sentence.** The Dandy stem says the company collects **30 percent** of receivables in the following month. The table shows 100 percent. Check it: 1,321,875 – 984,375 =

337,500, which is all of October's credit sales. The table wins.

### 3. Type 2 — Divisional growth to the fourth quarter

Three divisions or districts. Each has a **first quarter** figure and its own growth rate per quarter. Find fourth quarter sales.

**The one rule that decides the mark.** From the first quarter to the fourth quarter there are **THREE** growth steps, not four. Cube the multiplier.

$$Q4 = Q1 \times (1 + \text{growth})^3$$

#### GoldCom Corporation (also printed as Inky Pinky)

| Division                                        | First quarter | Growth | Working                 | Fourth quarter     |
|-------------------------------------------------|---------------|--------|-------------------------|--------------------|
| East Division                                   | \$520,000     | 2 %    | $520,000 \times 1.02^3$ | <b>\$551,828</b>   |
| West Division                                   | \$740,000     | 3 %    | $740,000 \times 1.03^3$ | <b>\$808,618</b>   |
| South Division                                  | \$340,000     | 5 %    | $340,000 \times 1.05^3$ | <b>\$393,593</b>   |
| <b>Total, the Q4 pro forma income statement</b> |               |        |                         | <b>\$1,754,039</b> |

#### Hokus Pokus Company Ltd (also printed as Akua Antwiwaah)

| Division                                        | First quarter | Growth | Working                 | Fourth quarter     |
|-------------------------------------------------|---------------|--------|-------------------------|--------------------|
| Kumasi District                                 | \$650,000     | 4 %    | $650,000 \times 1.04^3$ | <b>\$731,162</b>   |
| Accra District                                  | \$880,000     | 5 %    | $880,000 \times 1.05^3$ | <b>\$1,018,710</b> |
| Cape Coast District                             | \$420,000     | 10 %   | $420,000 \times 1.10^3$ | <b>\$559,020</b>   |
| <b>Total, the Q4 pro forma income statement</b> |               |        |                         | <b>\$2,308,892</b> |

#### Multipliers worth memorising

| Rate  | 2 %      | 3 %      | 4 %      | 5 %      | 10 %  |
|-------|----------|----------|----------|----------|-------|
| Cubed | 1.061208 | 1.092727 | 1.124864 | 1.157625 | 1.331 |

**Trap.** Raise the multiplier to the fourth power and you land exactly on one of the printed wrong options. That is what it is there for.

### 4. Type 3 — Cash budget with borrowing and interest

The hardest item, and the most predictable. Three months. The company holds a **cash cushion** before the interest payment. Interest runs at a fixed percent per month. Funds are borrowed and repaid **on the first day** of each month.

#### The seven steps

1. **Beginning cash balance** = last month's ending cash balance.
2. **Total cash available** = beginning balance + cash receipts.
3. **Total disbursements** = inventory purchases + selling and administrative expenses.
4. **Shortage or (surplus)** = disbursements – total cash available. Positive is a shortage, so borrow. Negative is a surplus, so repay.
5. **Financing.** Shortage: borrow = shortage + cushion. Surplus: repay = surplus – cushion.
6. **Interest** = rate × the **cumulative loan balance** after step 5. Track the running balance.
7. **Ending cash balance** = available – disbursements + borrowing – repayment – interest.

## Osagyefo Clothing Company Ltd — cushion GH¢40,000, interest 5 % a month

|                             | July              | August            | September         |
|-----------------------------|-------------------|-------------------|-------------------|
| Beginning cash balance      | GH¢42,500         | GH¢34,000         | GH¢31,000         |
| Cash receipts               | GH¢230,000        | GH¢486,000        | GH¢749,000        |
| <b>Total cash available</b> | <b>GH¢272,500</b> | <b>GH¢520,000</b> | <b>GH¢780,000</b> |
| Inventory purchases         | GH¢93,000         | GH¢409,100        | GH¢447,250        |
| Selling and admin. expenses | GH¢259,500        | GH¢130,900        | GH¢184,750        |
| <b>Total disbursements</b>  | <b>GH¢352,500</b> | <b>GH¢540,000</b> | <b>GH¢632,000</b> |
| Shortage / (surplus)        | GH¢80,000         | GH¢20,000         | (GH¢148,000)      |
| Cash cushion                | GH¢40,000         | GH¢40,000         | GH¢40,000         |
| Borrowing / (repayment)     | GH¢120,000        | GH¢60,000         | (GH¢108,000)      |
| Loan balance after          | GH¢120,000        | GH¢180,000        | GH¢72,000         |
| Interest                    | (GH¢6,000)        | (GH¢9,000)        | (GH¢3,600)        |
| <b>Ending cash balance</b>  | <b>GH¢34,000</b>  | <b>GH¢31,000</b>  | <b>GH¢36,400</b>  |

Total interest for the quarter: 6,000 + 9,000 + 3,600 = **GH¢18,600**

## Birdy-Birdy's Jewellery Shop — cushion GH¢9,000, interest 2 % a month

|                             | July              | August            | September         |
|-----------------------------|-------------------|-------------------|-------------------|
| Beginning cash balance      | GH¢7,500          | GH¢7,139          | GH¢7,214          |
| Cash receipts               | GH¢50,000         | GH¢180,000        | GH¢216,000        |
| <b>Total cash available</b> | <b>GH¢57,500</b>  | <b>GH¢187,139</b> | <b>GH¢223,214</b> |
| Inventory purchases         | GH¢110,000        | GH¢140,800        | GH¢164,736        |
| Selling and admin. expenses | GH¢31,560         | GH¢33,600         | GH¢35,280         |
| <b>Total disbursements</b>  | <b>GH¢141,560</b> | <b>GH¢174,400</b> | <b>GH¢200,016</b> |
| Shortage / (surplus)        | GH¢84,060         | (GH¢12,739)       | (GH¢23,198)       |
| Cash cushion                | GH¢9,000          | GH¢9,000          | GH¢9,000          |
| Borrowing / (repayment)     | GH¢93,060         | (GH¢3,739)        | (GH¢14,198)       |
| Loan balance after          | GH¢93,060         | GH¢89,321         | GH¢75,123         |
| Interest                    | (GH¢1,861)        | (GH¢1,786)        | (GH¢1,502)        |
| <b>Ending cash balance</b>  | <b>GH¢7,139</b>   | <b>GH¢7,214</b>   | <b>GH¢7,498</b>   |

Total interest for the quarter: 1,861 + 1,786 + 1,502 = **GH¢5,149**

**Where the marks are lost.** Interest is charged on the **running loan balance**, not on the new borrowing alone. In Osagyefo, August interest is 5 % of GH¢180,000, not of the GH¢60,000 just borrowed.

**Add the cushion when borrowing. Subtract it when repaying.** Getting that sign backwards is the second most common error.

## 5. Known errors in the printed papers

| Item | What is wrong | What to do |
|------|---------------|------------|
|------|---------------|------------|

|                                     |                                                                                                                                             |                                           |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|
| Kaneapa, AR on the Q4 balance sheet | The correct figure is December credit sales, \$54,450. It is not printed as an option. The official key marks \$78,650, the December total. | Pick the December figure that is offered. |
| Dandy Electronics                   | The stem says 30 percent collection. The table shows 100 percent.                                                                           | Trust the table.                          |
| Mercy's Bookshop                    | The stem gives a cushion of GH¢9,000. The table uses GH¢7,000.                                                                              | Use the figure in the table.              |
| Birdy-Birdy's, total interest       | Exact total is GH¢5,149. The option offered reads GH¢5,150.                                                                                 | Take the nearest printed figure.          |

## 6. The card. Read this last, in the hall.

**Type 1.** AR this month = **last month's credit sales**. October uses the opening balance. Cash sales are read off the table. AR on the balance sheet = **December credit sales**. Sales revenue = **sum of the three totals**.

**Type 2. Cube it.**  $Q4 = Q1 \times (1 + g)^3$ . Three steps, not four. Then add the divisions for the income statement figure.

**Type 3.** Available = opening + receipts. Gap = disbursements – available. Shortage: borrow gap + cushion. Surplus: repay surplus – cushion. Interest = rate × **running loan balance**. Ending = available – disbursements + borrowing – repayment – interest.

**Always.** Trust the table over the sentence. Work the first month fully and check you hit the printed ending balance before you trust the rest.